

2016.0 RANGE ROVER (LG), 303-01

ENGINE - TDV6 3.0L DIESEL - GEN 2/TDV6 3.0L DIESEL

RIGHT CYLINDER HEAD (G1715989)

REMOVAL AND INSTALLATION

12.29.05	CYLINDER HEAD - ASSEMBLY - RIGHT - RENEW	3000 CC, TDV6, DIESEL, WITH PARTICULATE FILTER, WITH DIESEL EXHAUST FLUID	11.3	USED WITHINS	+
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PART(S)

STEP	REPLACE PART / RENEW PART	PART NAME
Installation Step 2	Renew Part	Cylinder head gasket
Installation Step 2	Renew Part	Cylinder head bolts
Installation Step 11	Renew Part	Exhaust manifold gasket
Installation Step 13	Renew Part	Exhaust manifold cross-over pipe studs
Installation Step 14	Renew Part	Exhaust manifold cross-over pipe nuts

REMOVAL

NOTES:

- Removal steps in this procedure may contain installation details.
- Some variation in the illustrations may occur, but the essential information is always correct.

1. Disconnect the battery ground cable

Refer to: [Specifications](#) (414-00 Battery and Charging System - General Information, Specifications).

- 2.

WARNING:

Make sure to support the vehicle with axle stands.

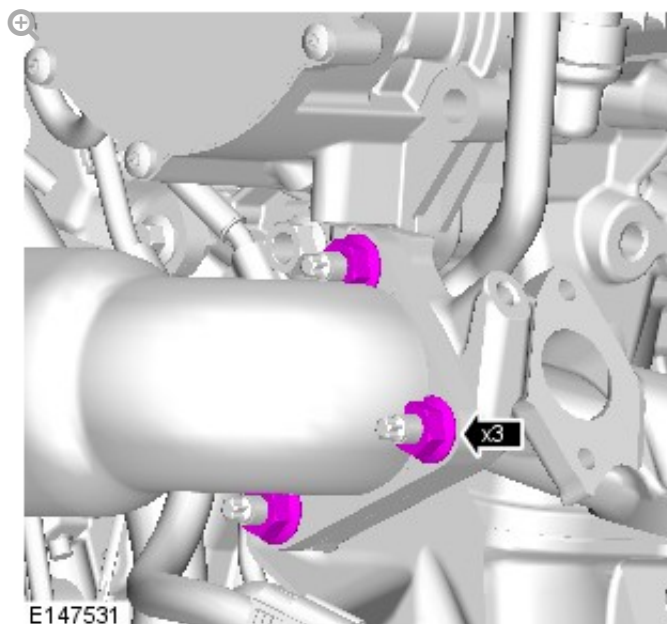
Raise and support the vehicle.

3. Refer to: [Right Turbocharger](#) (303-04B Fuel Charging and Controls - Turbocharger - TDV6 3.0L Diesel, Removal and Installation).

- 4.

CAUTION:

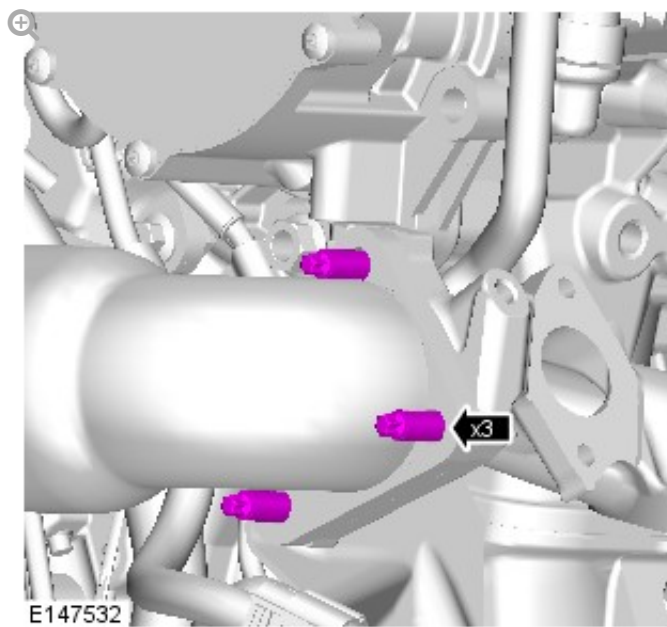
Discard the nuts.



5.

CAUTION:

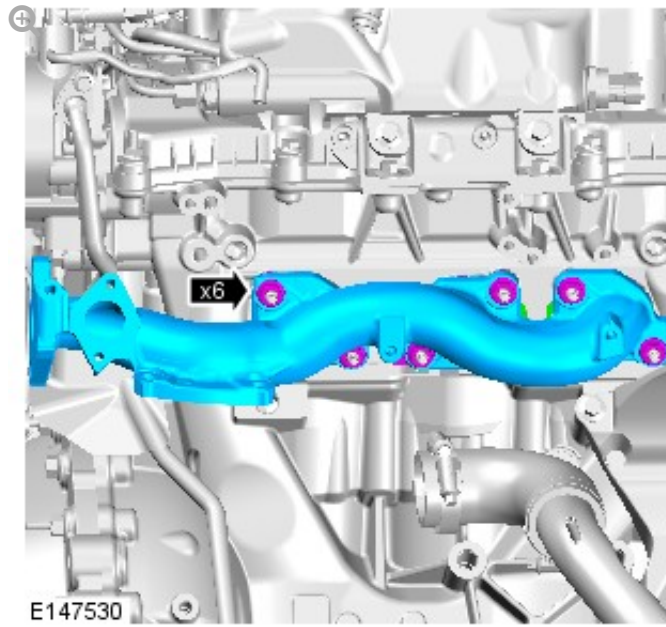
Discard the studs.



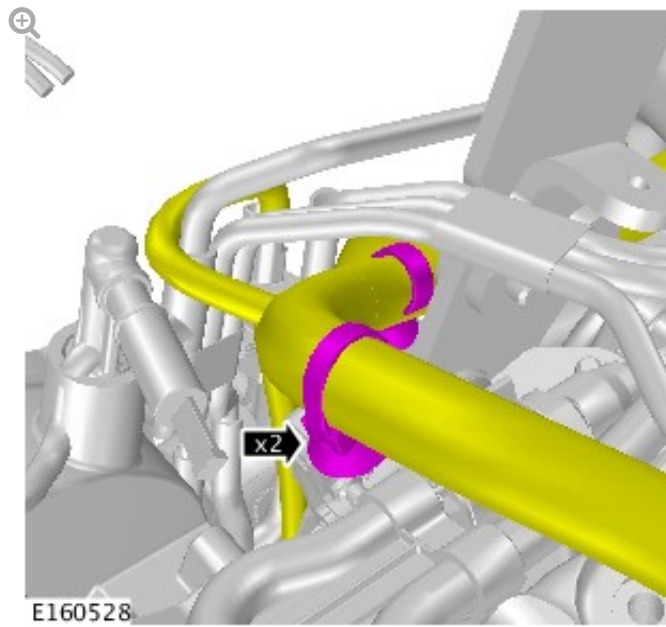
6.

NOTE:

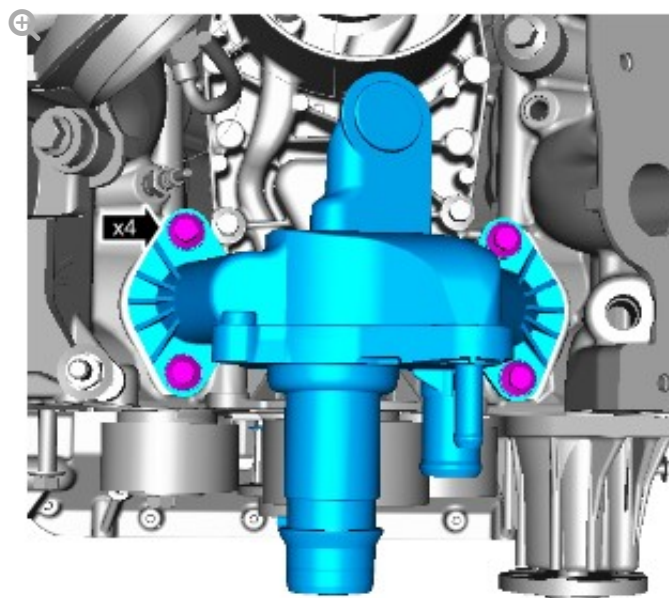
Discard the gasket.



7.

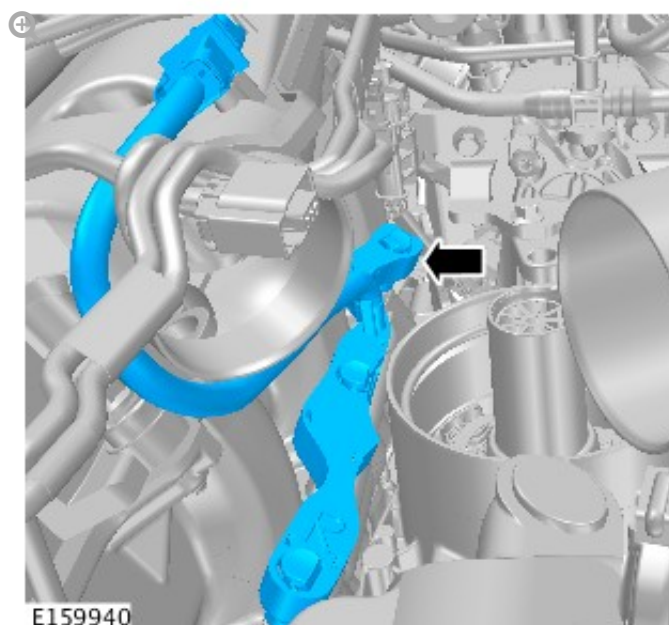


8.



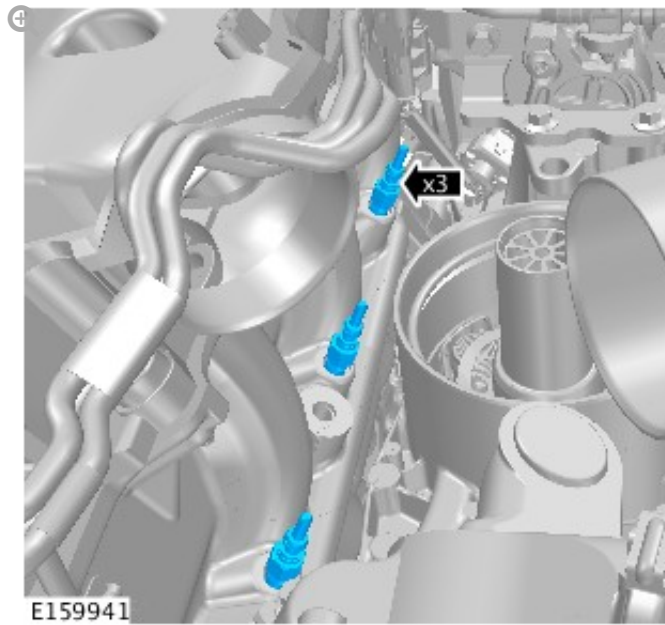
E116917

9.



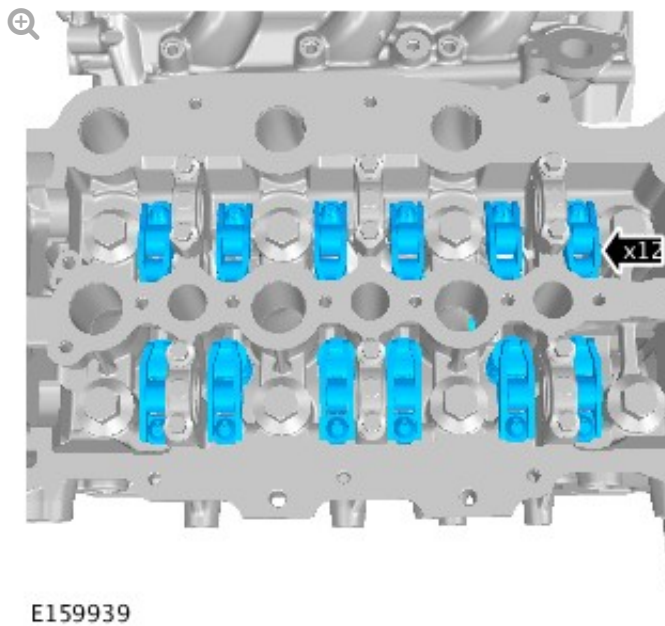
E159940

10.



11. Refer to: [Right Camshaft](#) (303-01A Engine - TDV6 3.0L Diesel - Gen 2/TDV6 3.0L Diesel, Removal and Installation).

12.



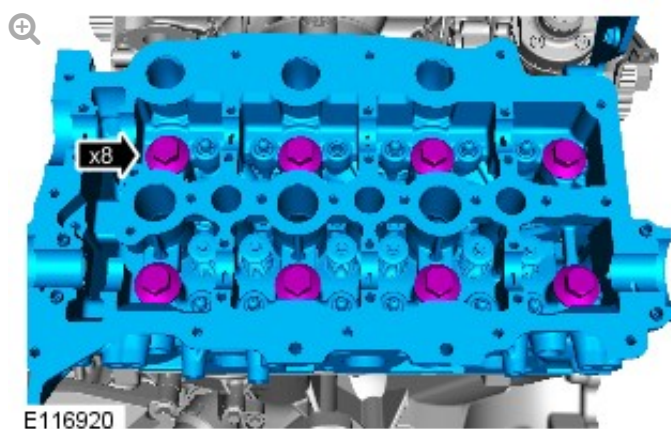
13.

CAUTION:

Discard the bolts.

NOTE:

Discard the gasket.

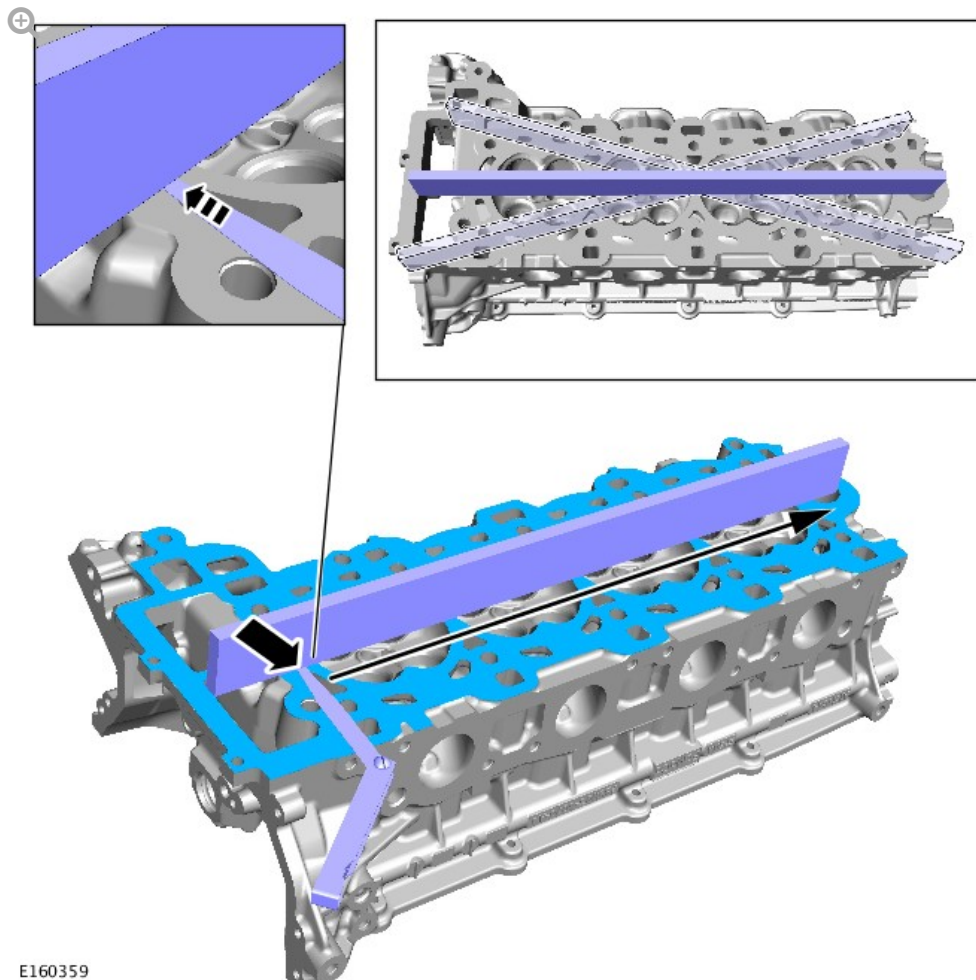


INSTALLATION

1.

CAUTIONS:

- Make sure that the mating faces are clean and free of foreign material.
- An acceptable flatness of the cylinder head is 0.2mm.



1. Clean and inspect the cylinder head and cylinder block.
2. Check the cylinder head face for distortion, across the center and from corner to corner.

Refer to: [Cylinder Head Distortion](#) (303-00 Engine System - General Information, General Procedures).

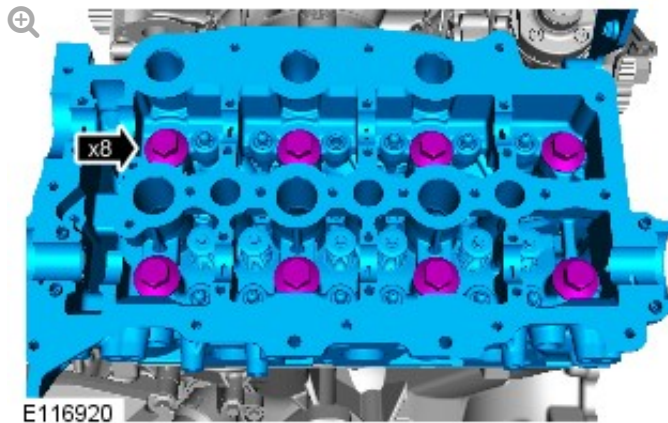
Refer to: [Cylinder Head Gasket Selection - TDV6 3.0L Diesel](#) (303-00 Engine System - General Information, General Procedures).

CAUTIONS:

- Only tighten the bolts finger-tight at this stage.
- Make sure that new bolts are installed.
- Use care when installing the cylinder head. Damage to the cylinder block, cylinder head or cylinder head gasket may result.
- The head gasket must be installed over the cylinder block dowels.

NOTES:

- Install a new gasket.
- No additional lubrication to the cylinder head bolts is required.



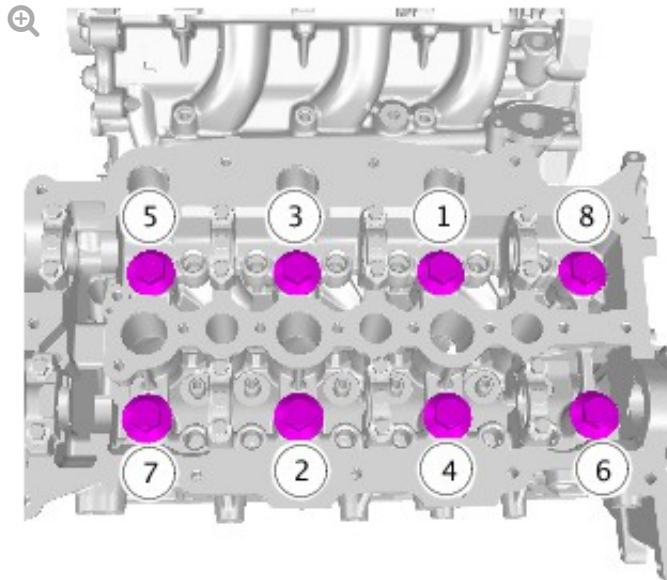
Renew Part: [Cylinder head gasket.](#)

Renew Part: [Cylinder head bolts.](#)

3.

NOTE:

Tighten the bolts in the indicated sequence.



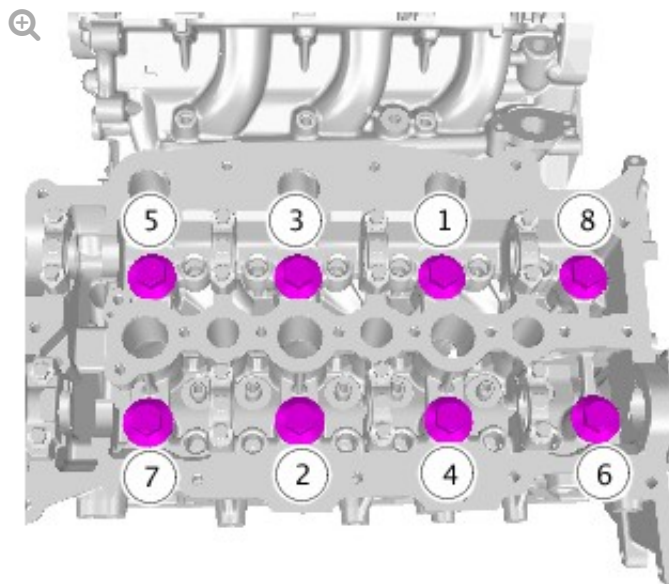
E159942

- *Torque: 80 Nm*

4.

NOTE:

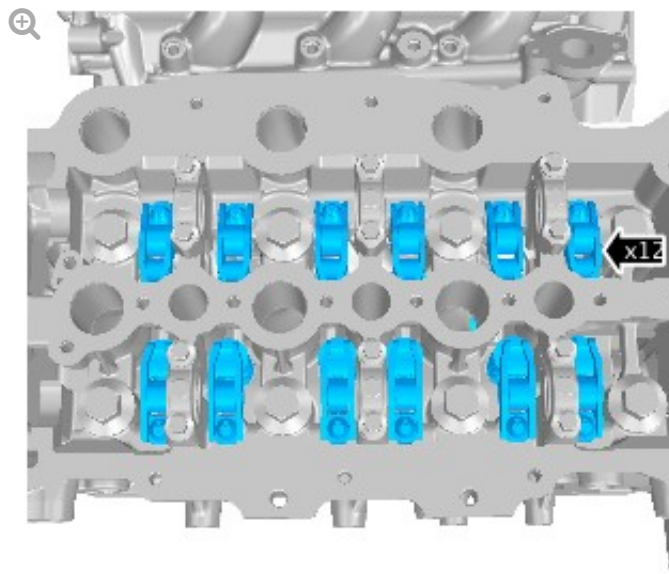
Tighten the bolts in the indicated sequence.



E159942

- *Torque: 180°*

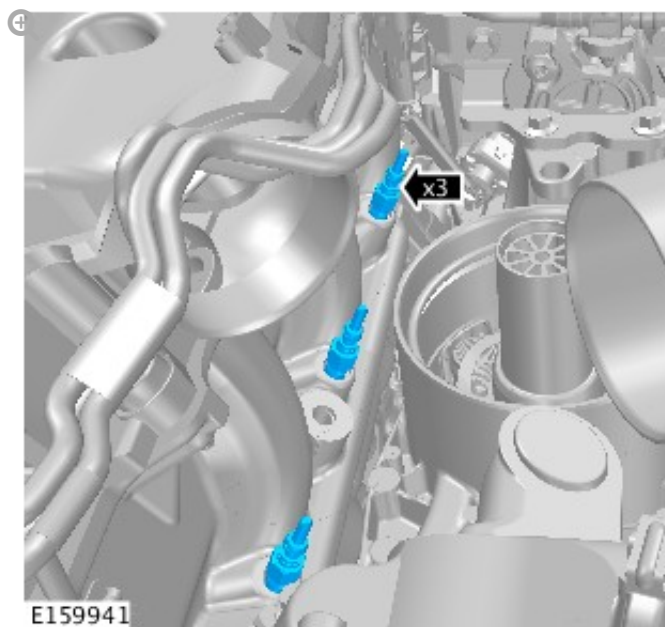
5.



E159939

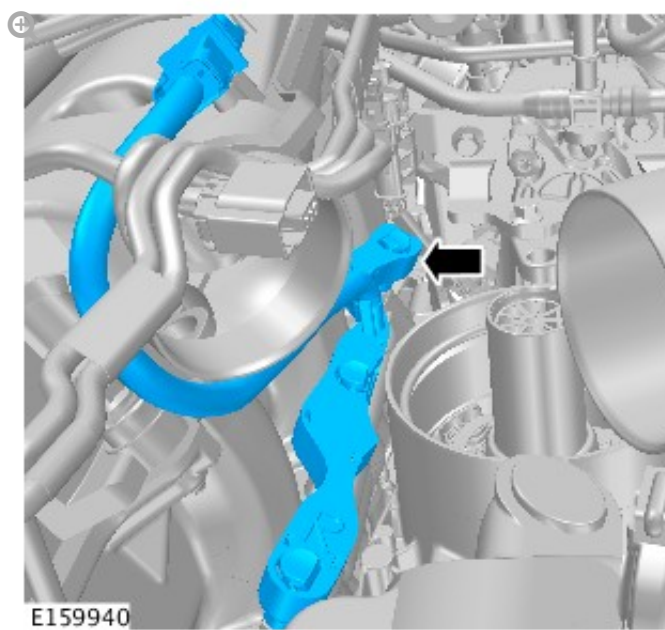
6. Refer to: [Right Camshaft](#) (303-01A Engine - TDV6 3.0L Diesel - Gen 2/TDV6 3.0L Diesel, Removal and Installation).

7.

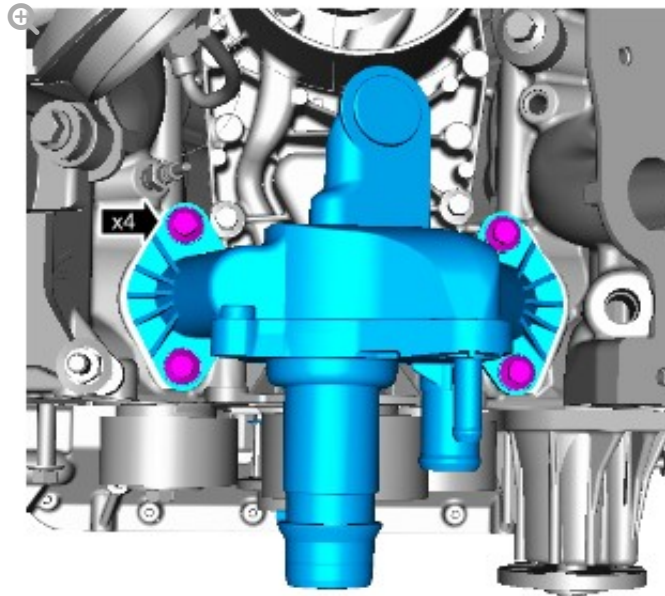


Torque: 10 Nm

8.



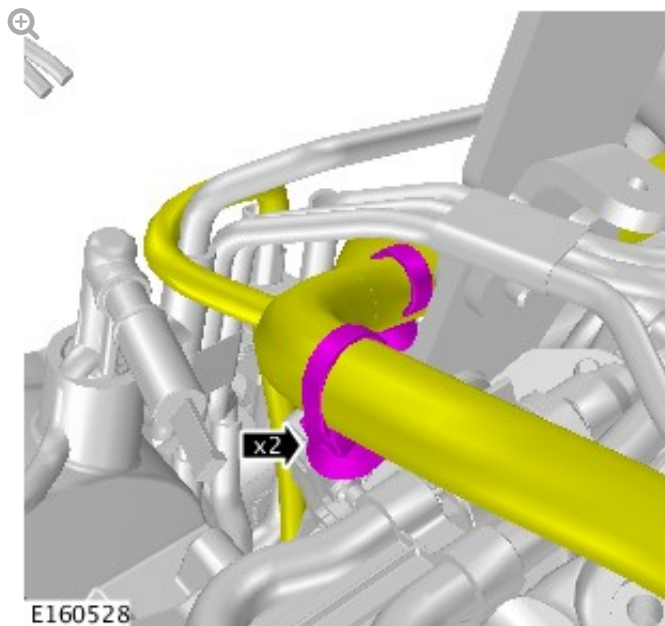
9.



E116917

Torque: 10 Nm

10.



E160528

11.

WARNING:

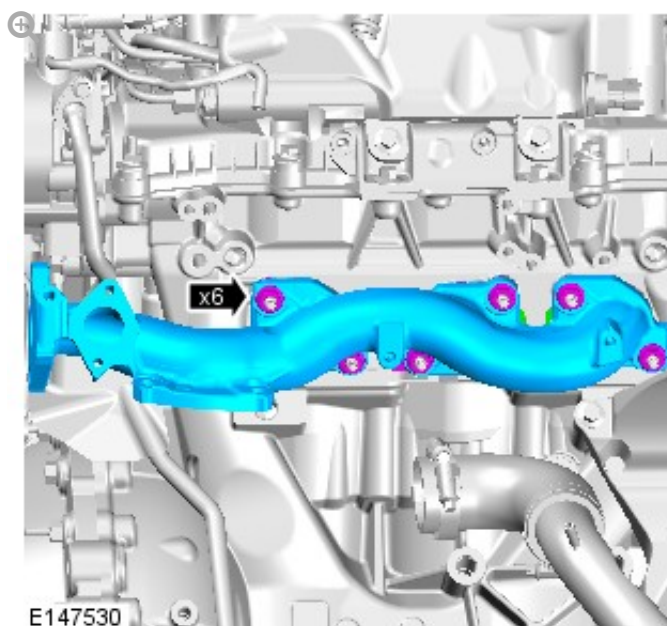
Make sure that new nuts are installed.

CAUTION:

Install the nuts finger tight before final tightening.

NOTE:

Install a new gasket.

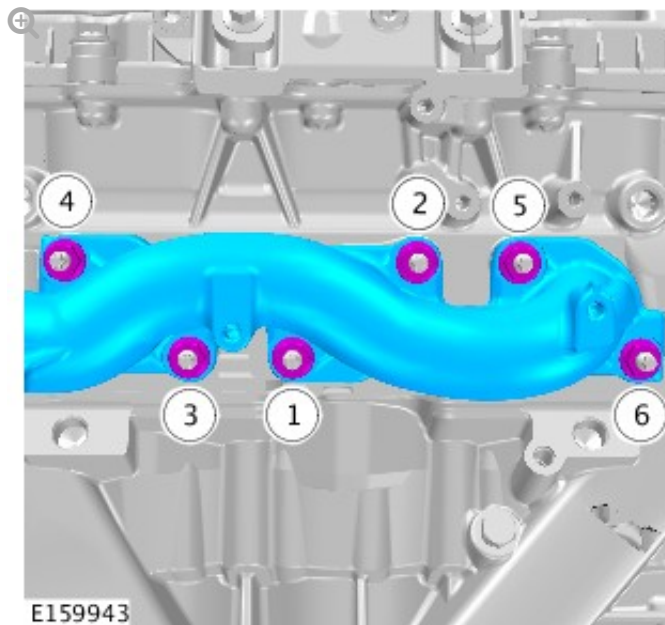


Renew Part: *Exhaust manifold gasket.*

12.

NOTE:

Tighten the bolts in the indicated sequence.

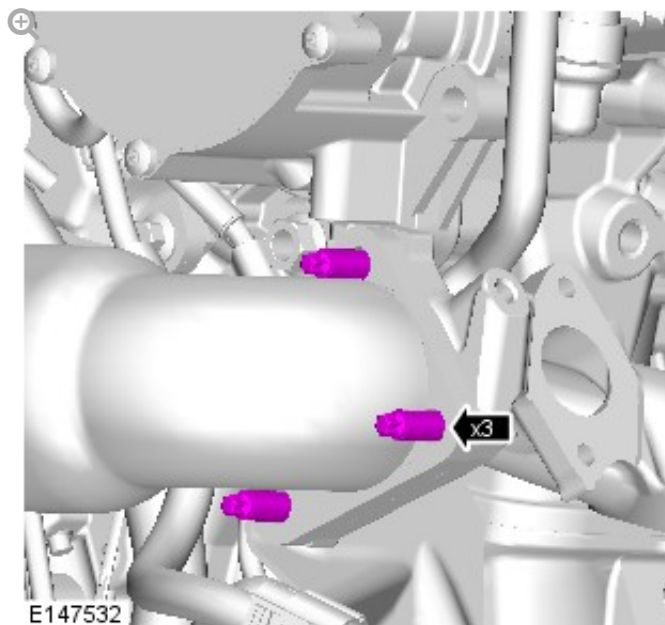


Torque: 28 Nm

13.

CAUTION:

Install new studs.



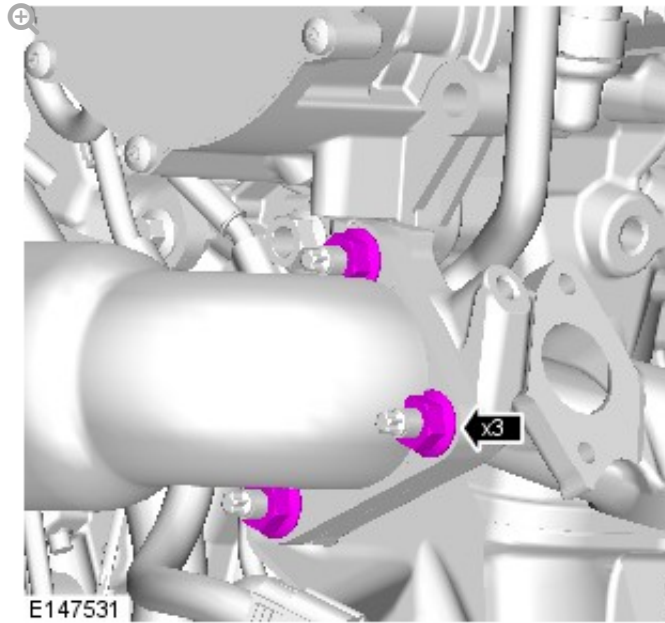
Renew Part: *Exhaust manifold cross-over pipe studs.*

Torque: 13 Nm

14.

WARNING:

Make sure that new nuts are installed.



Renew Part: [Exhaust manifold cross-over pipe nuts](#).

Torque: **24 Nm**

15. Refer to: [Right Turbocharger](#) (303-04B Fuel Charging and Controls - Turbocharger - TDV6 3.0L Diesel, Removal and Installation).
16. Connect the battery ground cable
Refer to: [Specifications](#) (414-00 Battery and Charging System - General Information, Specifications).